Semantic Segmentation of Land Cover in Urban Areas by Fusing Multisource Satellite Image Time Series

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***Abstract*— Due to the complex and highly heterogeneous land cover in urban areas, the single-temporal pixel-wise and parcel-wise classification cannot realize high-precision recognition of ground objects. Semantic segmentation of satellite image time series (SITS), can distinguish objects with similar spectral reflection and temporal evolution. However, optical SITS have problems of uneven time–frequency distribution and incomplete, which makes it impossible to directly use existing models to carry out time-series semantic segmentation. This study proposes a semantic segmentation network that combines optical and radar SITS, named multisource temporal attention (TA) fusion-based temporal–spatial transformer (MTAF-TST), to achieve high-precision land cover classification in urban areas. First, MTAF-TST uses the transformer spatial semantic segmentation module to extract the spatial context information of ground objects to realize pixel-level land cover classification, which relieves the salt-and-pepper phenomenon that is easy to occur in traditional pixel-by-pixel classification in complex scenes. Second, MTAF-TST uses the transformer time feature extraction module to mine long-range time-dependent and high-level semantic information, overcoming the drawbacks of traditional convolutional and recurrent neural networks that cannot mine long-range time-dependent features of SITS. Finally, MTAF-TST uses a multisource TA fusion module to fuse the depth features of optical and radar SITS, which overcomes the shortcomings of traditional direct feature stitching methods that cannot make full use of time-correlated features, achieving high-precision land cover classification. The experimental results show that the MTAF-TST can realize the complementarity of radar and optical SITS in terms of timing integrity, color, and texture, and effectively improve the accuracy of SITS classification.**

***Index Terms*— Land cover, multisource data fusion, satellite image time series (SITS), semantic segmentation, temporal–spatial transformer (TST).**

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I. Introduction

LAND cover changes reflect the underlying processes of nature, ecological environment, and human society [1], and affect climate and biodiversity on a global scale. Big Earth observation data is an important source of land cover change monitoring [2]. With the development of Earth observation technology, more and more high spatial and high temporal resolution optical and radar satellite image time series (SITS) are accumulated [3], which provide a solid data foundation for monitoring the dynamic changes of land [4], forest [5], ocean [6], and phenological [7]. However, the complex and unbalanced distribution of impervious surfaces, green land, water, and unused land in urban areas presents a high degree of heterogeneity, which poses a huge challenge for land cover classification [8]. The traditional single-temporal pixel-wise and parcel-wise classification cannot use the spatiotemporal context information of ground objects, often leading to imprecise classification results [9].

SITS contains the reflection characteristics of land cover under different phenological conditions, which can help improve and enhance the performance of classification models [10]. Optical remote sensing images, especially multispectral and hyperspectral images, can obtain the reflectance values of the same object in different spectral bands, as well as other important attribute information like the color and texture of the ground objects. Therefore, it is the main data source for land cover time series classification [11]. However, optical remote sensing is greatly affected by external factors like weather and illumination, which seriously affect the quality of optical time series. Synthetic aperture radar (SAR), which belongs to the category of active remote sensing, can provide 24-h all-weather Earth observation, to obtain even and complete time series Earth observation datasets. However, SAR cannot capture the rich color information of ground objects. In addition, SAR SITS may be often affected by speckle noise [12], which reduces the accuracy of land cover classification. Therefore, fusing the rich color and texture information of optical time series, and the even, complete,

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and multiscale time dependence of radar time series, may be the key to improving the accuracy of land cover classification.

However, traditional SITS classification models mostly use optical [13] or SAR [14] data sources alone, and most of them take single-pixel time series as the basic analysis unit [15], which means that the traditional methods cannot mine the spatial context information and long-range time-dependent of SITS. That is to say, the traditional methods cannot make full use of the high-level semantic information of remote sensing images to improve the land cover classification accuracy. In addition, traditional optical and radar timing fusion methods mostly use direct splicing mode, which cannot make full use of time-dependent features between different data sources.

In order to make full use of the spatial-temporal synergy and the complementary advantages of multisource remote sensing images, this study proposes a multisource temporal attention (TA) fusion-based temporal–spatial transformer (MTAF-TST) semantic segmentation framework to carry out semantic segmentation of land cover using optical and SAR SITS. The specific contributions are as follows.

1) Using the transformer spatial semantic segmentation module to extract the spatial context information of ground objects to realize pixel-level land cover classification, relieves the salt-and-pepper phenomenon that is easy to occur in traditional pixel-by-pixel classification in complex scenes.
2) Using the transformer spatiotemporal feature extraction module to mine long-range time-dependent and high-level spatial semantic information, overcoming the drawbacks of traditional convolutional and recurrent neural networks that cannot mine long-range dependent features of SITS.
3) Using a multisource TA fusion module to fuse the depth features of optical and radar time series, overcoming the shortcomings of traditional direct feature stitching methods that cannot make full use of time-correlated features.

II. Related Work

Semantic segmentation of SITS is to achieve pixel-by-pixel classification of remote sensing images by mining the spatiotemporal context and advanced semantic information of land cover. Therefore, we review the literature from three aspects: SITS spatiotemporal feature extraction, multisource data fusion, and semantic segmentation of land cover, summarize the problems existing in current research, and propose our solutions.

A. SITS Spatiotemporal Feature Extraction

In some early studies, researchers extracted the features of each image and then used machine learning (such as support vector machine [16] and random forest [17]) to complete the SITS land cover classification. However, these method has two shortcomings. One is that it regards SITS as a collection of images, ignoring the time series correlation between different time [18], and the other is that for traditional machine-learning algorithms, the splicing of multitemporal features often leads to the problem of "cures of dimensionality" [19]. Other studies extract phenological information from the spectral trajectories of each pixel time series, such as spectral statistics [20] and phenological parameters [21]. However, these hand-extracted features provide limited discriminative information and are not robust.

In recent years, people have used complex networks to model SITS, which can automatically extract powerful feature discrimination information without the need for manual feature selection. Pelletier et al. [22] introduced temporal convolutional neural networks (TempCNNs) into crop classification, using 1-D convolution operations to extract temporal features in the temporal dimension. Wang et al. [23] proposed that using a bidirectional long–short-term memory network (Bi-LSTM) to consider both the forward and reverse temporal dependencies of a time series, and confirmed that the Bi-LSTM is superior to a convolutional neural network (CNN) or a long–short-term memory (LSTM) model. Rußwurm and Körner [24] conducted detailed experiments using several commonly used deep-learning models and showed that transformers and LSTM outperform TempCNNs when classifying unprocessed SITS data. Yuan and Lin [25] found that transformers have a stronger anti-overfitting ability when the training samples are limited.

In addition to applying deep learning to extract temporal dependencies on single-pixel time series, some studies have begun to use 3D-CNNs [26] or ConvLSTM [27] to simultaneously extract temporal and spatial features of SITS. This method models both spatial and temporal dimensions and extracts spatiotemporal features together, which improves the expressiveness of features. Interdonato [18] designed a network structure with two branches, one to extract the temporal dependencies of SITS and another to extract the spatial dependencies, and spliced them into spatiotemporal features. This separate processing reduces the interaction between the distribution of temporal and spatial information.

In general, the current deep-learning methods mainly extract the features of SITS in the time domain or the spatiotemporal domain [28]. Therefore, in the temporal and spatial dimensions, mining spatiotemporal features with stronger expressiveness and robustness are the main way to improve the accuracy of land cover classification using SITS. However, it is difficult for deep-learning algorithms to simultaneously extract the long-range temporal and spatial dependencies of SITS. Therefore, in this article, we propose a self-attention mechanism-based temporal–spatial transformer module, TST, to extract long-range spatiotemporal dependencies.

B. Multisource Data Fusion

In view of the different capture capabilities of optical and radar sensors in terms of ground object color and texture characteristics, people often fuse multisource remote sensing data to improve the ground feature capture capability of a single sensor as much as possible. For example, Ren et al. [29] presented a locally optimized image segmentation fusion (LOISF) framework for hyperspectral images super-resolution reconstruction. In general, multisource data fusion includes three types: pixel-level fusion, feature-level

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fusion, and decision-level fusion [30]. In most cases, the pixel-level fusion strategy is adopted. That is, the optical and radar images are directly spliced into one image, and then traditional machine learning, such as a random forest [31], [32] or support vector machine [33], is used to carry out land cover classification. However, directly stitching multisource SITS will cause a disaster in high-dimensional feature space, which brings great challenges to traditional machine-learning algorithms. Therefore, people often use feature selection strategies to remove redundant features in high-dimensional features and retain features with low correlation, thereby improving the accuracy of land object classification [34].

Deep-learning models have powerful feature extraction capabilities and can automatically learn features that are conducive to improving classification accuracy, so they do not need to go through the feature selection stage. Liao et al. [35] directly used 1D-CNN to extract temporal features of optical and radar SITS. Both Ienco et al. [4] and Hosseiny et al. [36] used deep networks with two different branches to separately extract spatiotemporal features of optical and radar SITS, respectively, and then stitch the features of the two data sources at the feature level. Ofori-Ampofo et al. [37] compared three deep-learning-based data fusion methods, and the results show that when the object of interest is not fully represented, it is better to use a pixel-level or decision-level fusion method; otherwise, the three fusion methods are all possible. In addition, the authors argue that it is very meaningful to fuse the high-level semantic information of optical and SAR SITS when the optical data is very limited.

As such, the SITS fusion mostly adopts the direct stitching mode, such as direct stitching of image bands or latent representation. However, this fusion method does not use the time series properties of multisource images. Therefore, the study proposes a fusion method using multisource SITS association, MTAF, to improve the expression ability of fusion features.

C. Semantic Segmentation of Land Cover

The goal of semantic segmentation is to assign a semantic class to each pixel of the input image. Long et al. [38] first proposed fully convolutional networks (FCNs) to complete the task of semantic segmentation. In the field of biological image semantic segmentation, Ronneberger et al. [39] proposed a medical image segmentation network, U-Net, which splices the low-resolution image features from the encoder into the decoder, then restores the original image resolution scale through upsampling, and finally achieves pixel-level image classification. In addition, other commonly used semantic segmentation networks include pyramid scene parsing network (PSPNet) [40] and Deeplab v3 [41].

In the field of land cover semantic segmentation, Su et al. [42] proposed an improved U-Net network, which combines the advantages of dense convolutional network (DenseNet), U-Net, dilated convolution, and transposed convolution; improves the accuracy of land cover segmentation; and reduces the number of model parameters compared to the classic U-Net. Wang et al. [43] proposed a new semantic segmentation network, including the class feature attention mechanism (CFAM) and multiparallel atrous spatial pyramid pooling (MASPP), two modules in which the CFAM can enhance the ability of the semantic segmentation network to understand information between different types and help learn different types of semantic information. The MASPP can also help learn multiscale spatial information. Xie et al. [44] proposed a semantic and spatial-spectral feature fusion transformer (S3FFT) network to mine shallow spatial-spectral features and then fuse the semantic features extracted by the transformer-based module to complete land cover semantic segmentation. However, the spatial and spectral features extracted by Xie et al. [44] proposed transformer-based semantic segmentation methods, interact with each other, which to some extent affects the improvement of classification accuracy.

There are few studies on semantic segmentation with time series remote sensing images. Mehra et al. [45] compared six deep-learning architectures (pyramid scene parsing, U-Net, DeepLab v3+, path aggregation network, encoder–decoder network, and feature pyramid network) using a multitemporal SAR time series for land cover semantic segmentation, and the results demonstrate the feasibility of classifying SAR SITS using a deep-learning-based semantic segmentation network.

Through the above review of SITS spatiotemporal feature extraction, multisource data fusion, and semantic segmentation of land cover, we propose the MTAF-TST network, to extract and fully fuse long-range spatiotemporal dependencies and advanced semantic information of SITS.

III. Methodology

The MTAF-TST network mainly consists of two parts, which are the data fusion module based on MTAF and the spatiotemporal information extraction module based on TST. The former mainly realizes information fusion between Sentinel-1 and Sentinel-2 SITS, making full use of the complementarity of optical and SAR data; the latter is mainly used to extract spatiotemporal features from SITS and achieve semantic segmentation.

A. Overall Architecture of MTAF-TST

The MTAF-TST network has three branches. The upper and lower branches have the same network structure, with one to extract the spatiotemporal features of optical time series and the other to extract the spatiotemporal features of SAR time series [46]. The middle branch first uses MTAF to fuse optical and SAR data and then uses TST to extract the spatiotemporal features of the fused data. Finally, the spatiotemporal features extracted by all three branches are merged to identify the land cover type. The overall architecture of MTAF-TST is as shown in Fig. 1.

B. Multisource Information Fusion Based on MTAF

Traditional optical and SAR time series fusions mostly use direct splicing or simple mathematical operations [37], [47]. In the calculation process, these two kinds of data are actually completely independent, and the complementary advantages of the two kinds of data are not fully utilized. Therefore, we propose an MTAF to realize that each time point of one time series pays attention to all time points of the other time series with different weights, as shown in Fig. 2.

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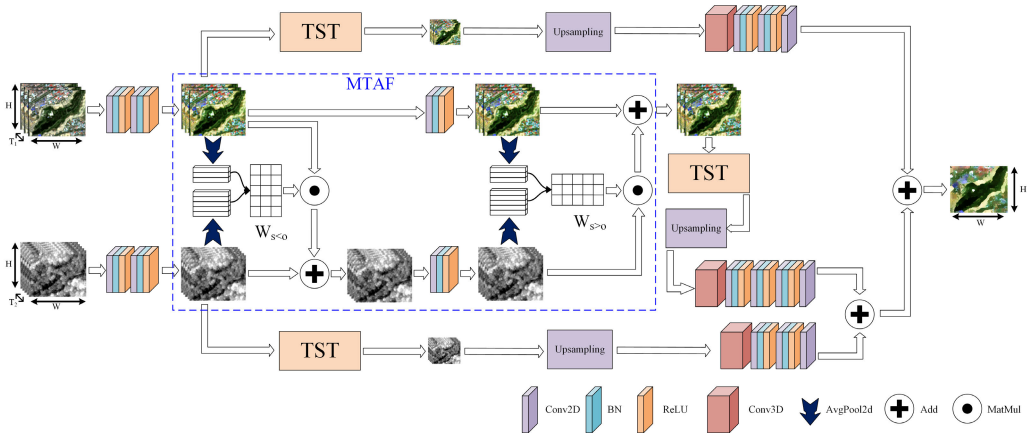


Fig. 1. Overall architecture of MTAF-TST. Conv2D and Conv3D represent 2-D and 3-D convolution layers, respectively, BN is the batch normalization layer, and the ReLU is the activation function. AvgPool2d represents the 2-D average pooling operation, and Add and MatMul represent addition and matrix multiplication, respectively. For the upper and lower branches, the input sequence first goes through two convolution blocks to extract shallow spatial features, then the TST module is used to successively mine temporal dependencies and more abstract spatial information, and finally, a series of deconvolutional and convolutional blocks are used for image size restoration and land cover classification. The middle branch is similar to the upper and lower branches, but MTAF is used to fuse optical and SAR data before TST.

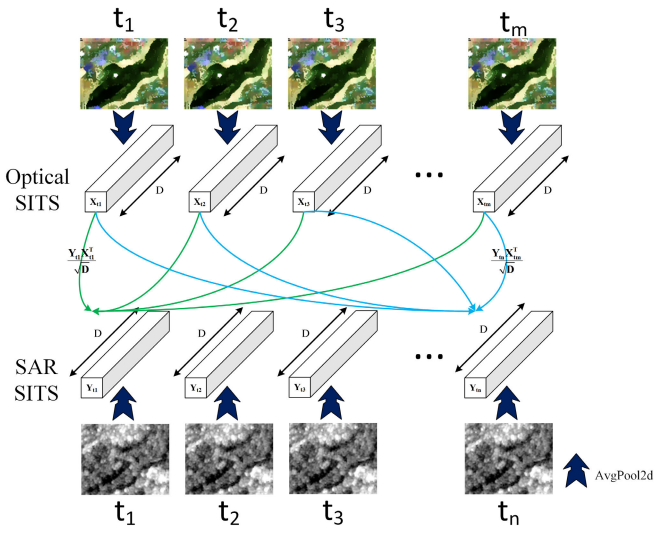


Fig. 2. Design idea of MTAF.

First, we found that the spectral characteristics captured by optical images are the most important information to distinguish the types of land cover, SAR SITS are relatively dense and complete, but they cannot capture the rich spectral characteristics of land cover. So we first let the SAR SITS pay attention to the optical SITS. If the SAR SITS pay attention to the optical SITS, the rich spectral features of the optical SITS can be introduced to enrich the information of the SAR SITS.

Second, the dense and complete time series evolution characteristics of land cover captured by a SAR time series can further improve the classification accuracy of land cover. Therefore, we next let the optical SITS pay attention to the SAR SITS to introduce the complete time series features into the optical SITS and enhance its feature expression ability.

Fig. 2 shows how the SAR SITS pays attention to the optical SITS. Equations (1) and (2) are the calculation methods for SAR pay attention to optical sequences

$$\hat{Y}_{ti} = Y_{ti} + \sum_{j}^{T} W_{s<o}[i, j] X_{tj} \tag{1}$$

$$W_{s<o}[i, j] = \text{Softmax}\left(\frac{Y_{ti} X_{tj}^{T}}{\sqrt{D}}\right) \tag{2}$$

where Y_{ti} denotes the representative vector of the SAR feature map at time ti and X_{tj} denotes the representative vector of the optical feature map at time tj. D represents the hidden layer dimension. Softmax$((Y_{ti} X_{tj}^{T}/\sqrt{D})$ represents the attention weight of the ti moment of SAR to the tj moment of optical SITS.

C. Spatiotemporal Information Extraction Based on TST

The main idea of the TST module is to use the self-attention mechanism to calculate the long-range dependencies between various moments or regions to fully extract spatiotemporal features. The TST module includes two parts, TA and spatial attention (SA), as shown in Fig. 3.

1) Temporal Attention: In the TA part, each time series pixel is processed individually, without considering any neighboring pixels. TA adopts the network structure of the traditional transformer [48] to process sequence data. The detailed neural network layers of TA are shown in Fig. 4. Assume that the input sequence is $X \in R^{T \times D}$, the position encoding is PE $\in R^{T \times D}$, and $X +$ PE constitutes the input of TA. Among them, T is the sequence length and D is the number of channels. The output sequence is X_{out} [see (3)]. After TA, X_{out} and X maintain the same shape

$$X_{\text{out}} = \text{FFN}(\text{Attention}(X + \text{PE})). \tag{3}$$

The Attention in (3) represents the self-attention mechanism

$$\text{Attention}(X) = \text{LN}\left(\text{Softmax}\left(\frac{Q K^{T}}{\sqrt{D}}\right) \cdot V + V\right), \quad \text{where}$$
$$Q = W_Q X, \quad K = W_K X, \quad V = W_V X. \tag{4}$$

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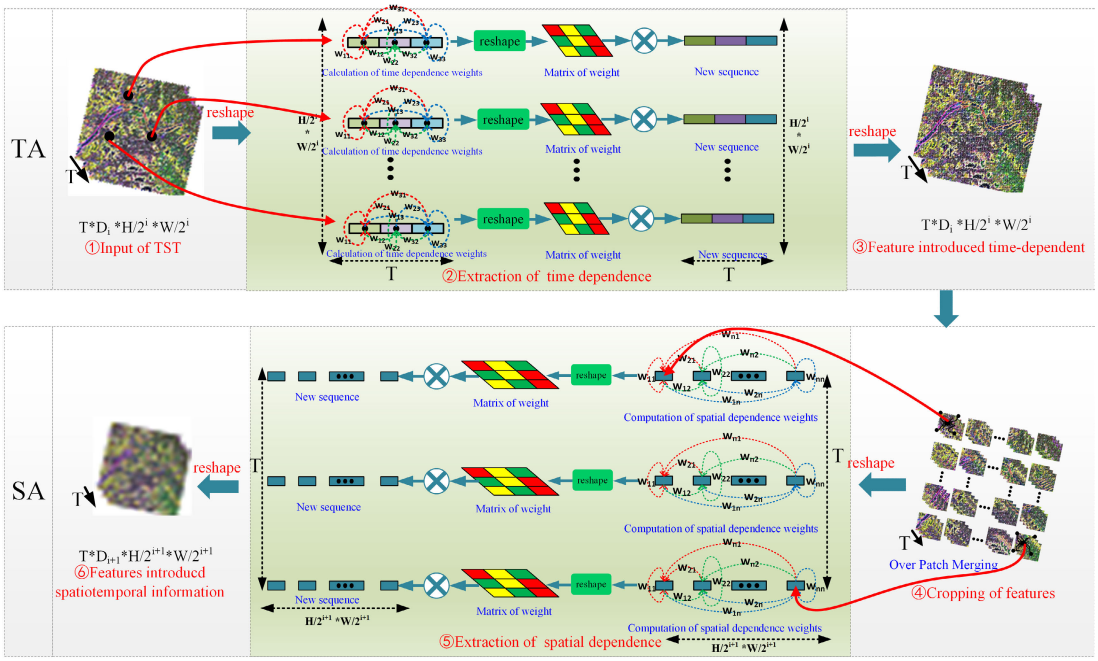


Fig. 3. Structure of TST.

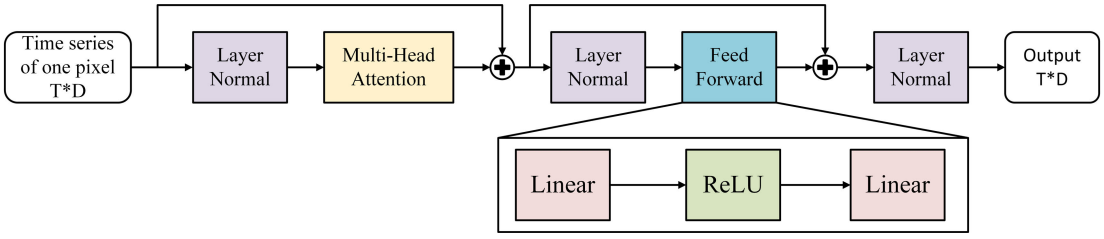


Fig. 4. Detailed network layers of TA.

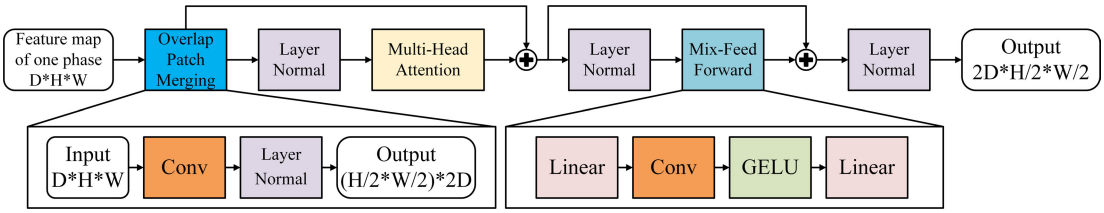


Fig. 5. Detailed network layers of TST.

And the feed-forward network (FFN) in (3) represents the feedforward neural network in transformer

$$\mathrm{FFN}(X) = \mathrm{Linear}(\mathrm{ReLU}(\mathrm{Linear}(X))). \tag{5}$$

Equation (6) is the formula of PE, where pos represents the position in the time series, $2i$ and $2i+1$ represent the even and odd positions in the embedding dimension

$$\begin{aligned} \mathrm{PE}(\mathrm{pos}, 2i) &= \sin(\mathrm{pos}/10000^{2i/D}) \\ \mathrm{PE}(\mathrm{pos}, 2i+1) &= \cos(\mathrm{pos}/10000^{2i/D}). \end{aligned} \tag{6}$$

2) Spatial Attention: In the SA part, the image of each moment is processed separately, regardless of the front and back phase data. Suppose the shape of the SA input feature data is $X \in R^{D\times H\times W}$, where D represents the number of channels and H and W represent the height and width, respectively.

First, SA embeds a neighboring pixel area as a vector; then, following the vision transformer (ViT) mode, the image expanded into a sequence, and the self-attention mechanism is used to calculate the spatial dependencies between different areas; finally, the sequence will be reexpanded into a 2-D feature map of shape $H/2 \times W/2$. Note that the original input features of shape $H \times W$ will be resized to $H/2 \times W/2$ due to the embedding of multiple pixels into a vector.

The detailed neural network layers are shown in Fig. 5. SA is similar to TA, but there are two differences. The one difference is that SA adds the overlap patch merging (OPM) module, which uses convolution to embed adjacent pixels into a token embedding

$$X_{\mathrm{sequence}} = \mathrm{Expand}(\mathrm{LayerNorm}(\mathrm{Conv2d}(X))) \tag{7}$$

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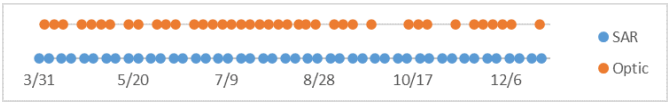


Fig. 6. Date distribution map of Ili dataset.

where $X \in R^{D\times H\times W}$ represents the input 2-D image, Expand represents expanding the 2-D matrix, and $X_{\text{sequence}} \in R^{(HW/4)\times D}$ represents the expanded sequence.

Another difference is the FFN. SA uses the mix-feed forward which is an improved version of FFN. The main difference is SA introduces a 3×3 convolutional layer in FFN and replaces rectified linear unit (ReLU) with Gaussian error linear unit (GELU)

$$\begin{aligned} X &= \text{Re_expand}(\text{Linear}(X_{\text{sequence}})) \\ \text{FFN}(X) &= \text{Linear}(\text{GELU}(\text{Expand}(\text{Conv2d}(X)))) \end{aligned} \tag{8}$$

where Re_expand means restoring the sequence to a 2-D image and GELU is the activation function.

D. Loss Function

MTAF-TST contains Branch-1, Branch-2, and Branch-3. The optimization goal of the model is to minimize the sum of the losses [see (9)] of each branch. The final classification result of the model only relies on the output of Branch-2. In our model, the role of Branch-1 and Branch-3 is to improve the features extracted from a certain data source, which in turn helps enhance the fused features in Branch-2

$$\text{loss} = \text{loss}_{\text{Branch}-1} + \text{loss}_{\text{Branch}-2} + \text{loss}_{\text{Branch}-3}. \tag{9}$$

IV. Experiment and Analysis

In this section, we will verify the superiority of the proposed model on the task of time series land cover classification through comparative experiments and ablation experiments. We selected two regions, Ili and Wuhan, and produced two time-series remote sensing image datasets. A detailed description of the datasets will be introduced later. In the comparison experiments, we compared the classification results of MTAF-TST and other deep-learning models. In the ablation experiments, we sequentially removed the TST-based spatiotemporal feature extraction module and the MTAF-based fusion module from the original model to verify the effectiveness of the two modules.

A. Experimental Dataset

1) Ili Dataset: Ili Kazakh Autonomous Prefecture is located in the Ili River Valley in the northern part of the Tianshan Mountains in the western part of the Xinjiang Uygur Autonomous Region, China. It belongs to the temperate continental climate zone and has few periods of rainy weather throughout the year. Therefore, the optical and SAR time series collected in Ili are suitable for making benchmark datasets. We selected 40 cloud-free optical images and 46 SAR images in Ili taken from April 2021 to December 2021, covering five small areas. The date distribution map of the optical and SAR time series is shown in Fig. 6.

TABLE I
Per Land Cover Statistics for the Ili Dataset

Label	Type	Pixels
0	Water	10316
1	Sparse vegetation(S-V)	10155
2	Cropland	14887
3	Forest	11725
4	Built-up	13787
5	Bare	12346
Total		73216

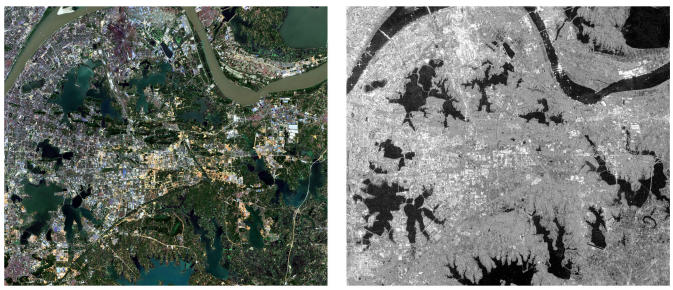


Fig. 7. Study area selected in Wuhan. (a) Optic. (b) SAR.

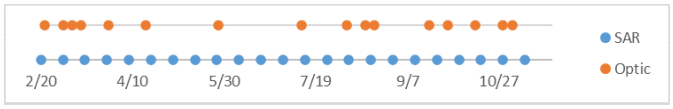


Fig. 8. Date distribution map of Wuhan dataset.

Both types of data came from Sentinel series satellites and were downloaded and preprocessed using Google Earth Engine (GEE). Optical data are Level-2A products, including four bands of B, G, R, and NIR. The necessary preprocessing operations (such as terrain correction, speckle filtering, and decibel conversion) were performed on the SAR data using GEE, which contains two bands of VV and VH. The spatial resolution was 10 m. The dataset contains six types of land cover which are water, sparse vegetation, cropland, forest, built-up, and bare. The number of manually labeled pixels for each land cover is shown in Table I.

2) Wuhan Dataset: Wuhan is located in the central part of China and belongs to the north subtropical monsoon climate zone, with abundant rainfall and sufficient heat all year round. The optical time series without cloud cover obtained in Wuhan is often incomplete. Therefore, the Sentinel-1 [49] and Sentinel-2 satellite time series collected from March 2020 to November 2020 in Wuhan (Fig. 7) were selected, where the length of the optical time series was 16 and the length of the SAR time series was 23. The date distribution map of the optical and SAR time series is shown in Fig. 8.

We referred to the European Space Agency's (ESA) 10-m land cover classification map and modified it according to the actual conditions of the study area. In the end, seven land cover categories were obtained, namely, tree cover, grassland, cropland, built-up, bare/sparse vegetation, permanent water

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TABLE II
PER LAND COVER STATISTICS FOR THE WUHAN DATASET

Label	Type	Pixels
0	Tree cover(TC)	2188018
1	Grassland(GL)	226053
2	Cropland(CL)	4515543
3	Built-up(BU)	3431174
4	Bare/sparse vegetation (B/SV)	1871292
5	Permanent water bodies (PWB)	3038929
6	Herbaceous wetland (HW)	138143
Total		15409152

bodies, and herbaceous wetlands. Fig. 7 shows the study area selected in Wuhan, and Table II counts the number of pixels in different categories.

3) Data Processing: We need to perform some processing operations to make the dataset. The image at one-time point is x_i, $x_i \in R^{D\times H\times W}$. $X = \{x_1, x_2, \ldots, x_T\}$ represents the remote sensing image time series. Where T is the timing length, D is the number of spectra, and H and W are the height and width of the image, respectively. Then, we calculate the mean and standard deviation of each spectrum in X to normalize each spectrum using z-score. After that, we use a sliding window to crop the image into individual patches p, $p \in R^{T\cdot D\times \mathrm{Win}\times \mathrm{Win}}$, where Win is the window size. Finally, 60% of the patches are randomly sampled as the training set, 20% of the patches are used as the verification set, and 20% of the patches are used as the test set. In addition, we divided the dataset twice to avoid the contingency of classification results caused by data division. The datasets obtained by the two divisions are, respectively, named region_resample_one and region_resample_two. We use horizontal flipping and vertical flipping to increase the amount of train set, thereby improving the generalization ability of the model.

B. Experiment Settings

All the models in this article are implemented using the PyTorch framework, using Adam as the model optimizer, and using the cross-entropy loss function. During the training process, the learning rate is dynamically updated according to F1-Score, and the patience is set to 15. The initial learning rate is set to 0.0001, which becomes half of the original at each update. The maximum epoch is set to 100, and the batch size is set to 4. The value of each indicator is the mean of region_resample_one and region_resample_two.

C. Classification Results

For the comparison model, we chose 2DU-Net + CLSTM [50] and the U-TAE-based dual-branch model [51]. The former removes the high-resolution imagery branch and only keeps the optical and radar imagery branches. The latter uses U-TAE as a spatiotemporal information encoder, and a two-layer convolutional neural net as a decoder, where the parameters of the spatiotemporal feature extraction modules of the optical branch and the radar image branch are not shared. Both models directly spliced the spatiotemporal information of optical and radar data for fusion.

1) Using Single Data Source: To provide a baseline, we first show the classification results of different models using only one kind of data.

As shown in Table III, all models are tested on the region_resample_one and region_resample_two, and the scores of each indicator are the average of the two experimental results. When only using optical or SAR time series, the 2DU-Net + CLSTM model is the worst in all indicators on both the Ili dataset and the Wuhan dataset. TST is better than U-TAE in all three indicators on the Ili dataset, but the performance of the two models on the Wuhan dataset is not much different. It can be seen that the proposed TST spatiotemporal extraction module is comparable to the current optimal U-TAE in extracting spatiotemporal information.

As can be seen in Fig. 9, the TST-based spatiotemporal information extraction module gets the highest classification accuracy on the five land covers of Ili dataset and the four land covers of Wuhan dataset.

2) Fusing Two Data Sources: Optical images have rich spectral information at each time point. However, due to the influence of cloud cover, the time distribution of the optical time series is not uniform, especially in the Wuhan dataset. That's because there is more cloudy and rainy weather in the southern region, and the frequency of cloud cover is higher.

However, deleting some time points will result in inconsistent time spans for adjacent moments in the sequence. Although the phenological evolution of land cover can still be obtained by using these nonuniformly distributed time-series data, the deletion operation is random. This randomness may lose some important timing information that we cannot perceive. Therefore, we need to supplement the missing information by introducing radar data.

(s1 + s2) indicates the model used the Sentinel 1 and Sentinel 2 data. Combining Tables III and IV, 2DU-Net + CLSTM showed a decrease in accuracy on the Ili dataset. However, overall accuracy (OA), F1-score, and mean intersection over union (mIoU) increased by 0.34%, 1.47%, and 1.36%, respectively, on the Wuhan dataset. The indicators of MTAF-TST and U-TAE are improved on both datasets. On the Ili dataset, compared with 2DU-Net + CLSTM and U-TAE, the accuracy of MTAF-TST has changed the most, and the three indicators have increased by 2.42%, 2.26%, and 3.09%, respectively. On the Wuhan dataset, MTAF-TST surpasses U-TAE(s1 + s2) and obtains the best classification results.

To verify the effectiveness of the MTAF-TST under smaller sample circumstances, more experiments with 30%, 45%, and 60% training samples have been conducted. However, the number of validation and testing sets remains unchanged to ensure consistency and reliability of evaluation standards. The precision evaluation results with different numbers of training samples are shown in Table IV. From Table IV, it can be seen that even with reduced training samples, MTAF-TST can still achieve the best classification results. In addition, the results of the Wuhan dataset indicate that increasing the training set is beneficial for improving the effectiveness of the model.

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TABLE III
CLASSIFICATION INDICATORS USING SENTINEL-2 DATA FROM ILI AND WUHAN

	Ili dataset			Wuhan dataset		
	OA	F1-score	mIoU	OA	F1-score	mIoU
2DU-Net+CLSTM(s1)	0.6550	0.6364	0.4923	0.7522	0.6331	0.4994
U-TAE(s1)	0.7064	0.6884	0.5428	0.7565	0.6426	0.5077
TST(s1)	0.7163	0.6981	0.5569	0.7579	0.6368	0.5042
2DU-Net+CLSTM(s2)	0.7573	0.7366	0.6078	0.8320	0.7269	0.6024
U-TAE(s2)	0.7981	0.7778	0.6618	**0.8340**	0.7359	0.6107
TST(s2)	**0.8163**	**0.8005**	**0.6870**	0.8339	**0.7394**	**0.6137**

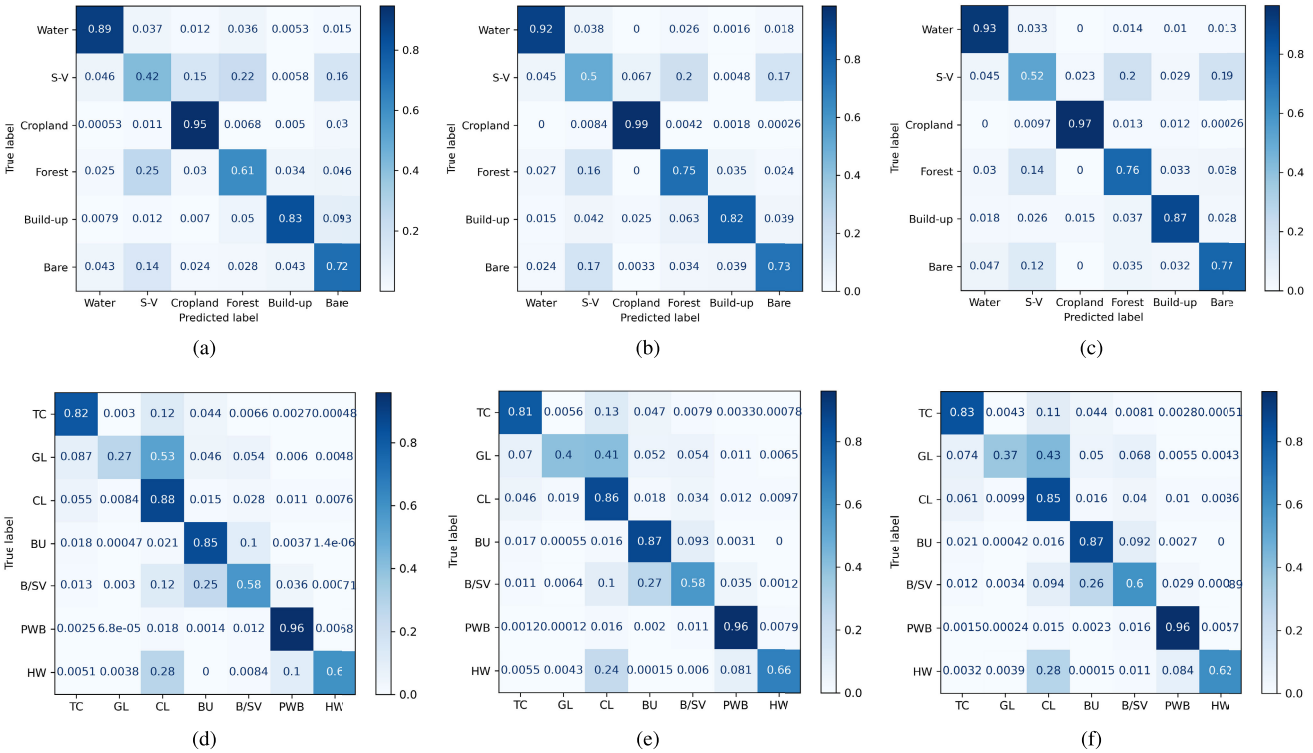


Fig. 9. Confusion matrix of the classification results of each model using only the Sentinel-2 data of the Ili dataset and Wuhan dataset. (a)–(c) Result of 2DU-Net + CLSTM(s2), U-TAE(s2), and TST(s2) on Ili dataset. (d)–(f) Result on Wuhan dataset.

As shown in Fig. 10, MTAF-TST is the best in all land covers on the Ili dataset, including sparse vegetation, crop, forest, and other land cover types that contain vegetation. On the Wuhan dataset, MTAF-TST achieves the best results on four land covers.

Fig. 11 shows the classification results of each model on the Ili dataset. On the whole, each model can correctly classify the land cover of most areas, but there are still some advantages and disadvantages in the details.

The part circled in Fig. 12 is the trees on both sides of the road. We found that only MTAF-TST can separate the trees on both sides of the road, resulting in two long strips of wooded areas. However, the 2DU-Net + CLSTM(s1 + s2) classification results are wrong. Although U-TAE(s1 + s2) is classified correctly, it confuses these two strips of woodland into one.

Fig. 13 circles many small ponds surrounded by grass. Water bodies and grass fields will appear very similar in the image, and it is difficult to distinguish the two with the naked eye. In Fig. 13, we can see that only MATF-TST correctly classifies these small ponds, while the other two models have a certain degree of confusion.

Fig. 14 circles a road. This road identified by MTAF-TST is the closest to the real label map, which is more continuous.

In Fig. 15, the classification results of each model are basically consistent with ground truth (GT). In the first region of Fig. 16, the built-up recognized by MTAF-TST and U-TAE(s1 + s2) is most similar to GT. However, 2DU-Net + CLSTM(s1 + s2) classifies Built-up worst. In the second region, MTAF-TST can identify a small water body in the upper right corner; however, neither 2DU-Net + CLSTM(s1 + s2) nor U-TAE(s1 + s2) can identify

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TABLE IV
PRECISION EVALUATION RESULTS WITH DIFFERENT NUMBER OF TRAINING SAMPLES USING SENTINEL-1 AND SENTINEL-2 DATA FROM ILI AND WUHAN

Training set	Model	Ili dataset			Wuhan dataset		
		OA	F1-score	mIoU	OA	F1-score	mIoU
30%	2DU-Net+CLSTM(s1+s2)	0.7142	0.6843	0.5515	0.8254	0.7250	0.5971
	U-TAE(s1+s2)	0.7906	0.7702	0.6503	0.8353	0.7452	0.6194
	MTAF-TST	0.8197	0.8021	0.6890	0.8385	0.7465	0.6218
45%	2DU-Net+CLSTM(s1+s2)	0.7412	0.7139	0.5898	0.8328	0.7342	0.6085
	U-TAE(s1+s2)	0.8034	0.7872	0.6715	0.8393	0.7480	0.6234
	MTAF-TST	0.8181	0.8015	0.6881	0.8409	0.7519	0.6278
60%	2DU-Net+CLSTM(s1+s2)	0.7354	0.7118	0.5817	0.8354	0.7416	0.6160
	U-TAE(s1+s2)	0.8082	0.7891	0.6767	0.8419	0.7485	0.6248
	MTAF-TST	**0.8405**	**0.8231**	**0.7179**	**0.8428**	**0.7522**	**0.6285**

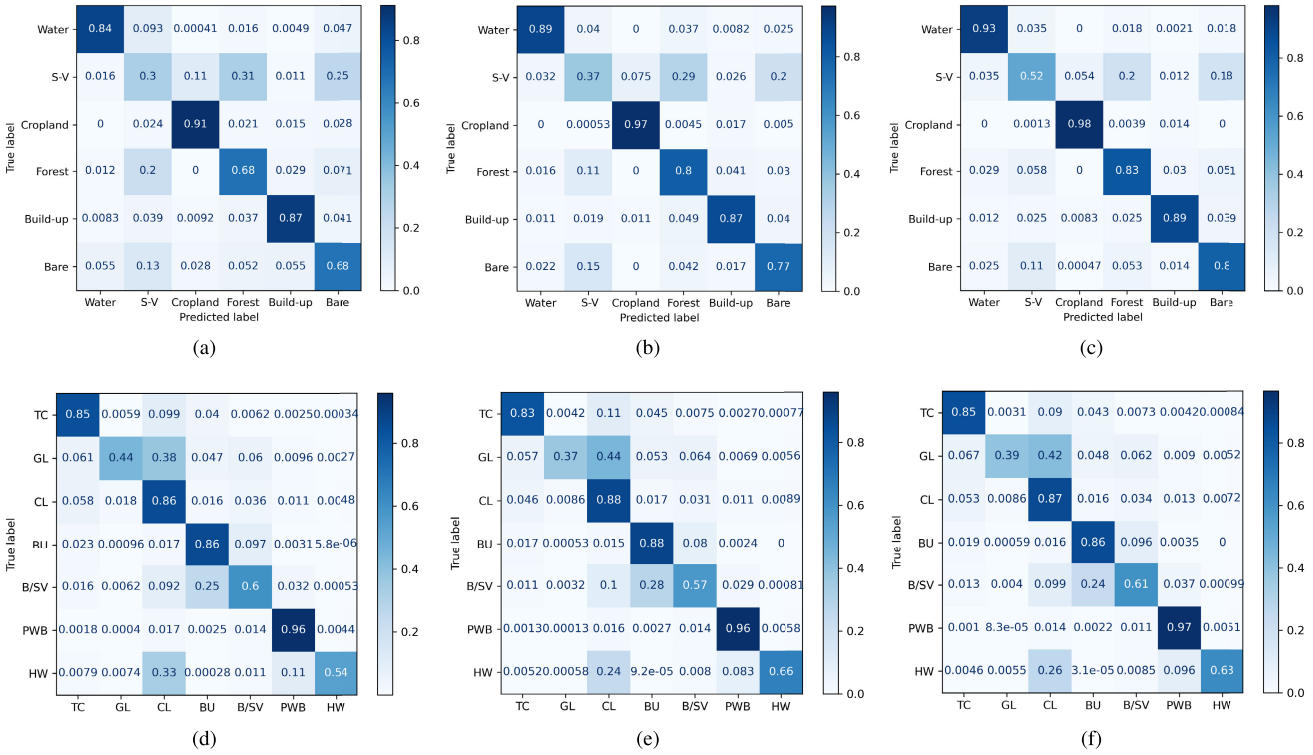


Fig. 10. Confusion matrix of the classification results of each model using Sentinel-1 and Sentinel-2 data of the Ili dataset and Wuhan dataset. (a)–(c) Result of 2DU-Net + CLSTM(s1 + s2), U-TAE(s1 + s2), and MTAF-TST on Ili dataset. (d)–(f) Result on Wuhan dataset.

it successfully. In the third region, MTAF-TST and GT are most consistent, especially for the identification of permanent water bodies. Among them, only MTAF-TST successfully identified the part of permanent water bodies in the inland area.

The t-SNE plots of the last layer features of MTAF-TST and comparison models are as shown in Fig. 17. It should be noted that all pixels in the Ili dataset are used for t-SNE rendering; however, only 10% of the pixels in the Wuhan dataset are used to draw t-SNE plots, as the Wuhan dataset is too large to visualize. It can be seen from Fig. 17 that, the clustering results of U-TAE(s1 + s2) and MTAF-TST are similar, but in the clustering results of 2DU-Net + CLSTM(s1 + s2), there is an obviously mixed crossover phenomenon. For example, in the Ili dataset, labels 1 and 3 are seriously mixed, and in the Wuhan dataset, labels 2 and 3 are interspersed with a large number of labels 0. Therefore, judging from the results of t-SNE, the results of U-TAE(s1 + s2) and MTAF-TST are equivalent, and 2DU-Net + CLSTM(s1 + s2) is the worst.

3) Ablation Experiments: MATF-TST mainly consists of two parts, which are the data fusion module based on MTAF and the spatiotemporal information extraction module

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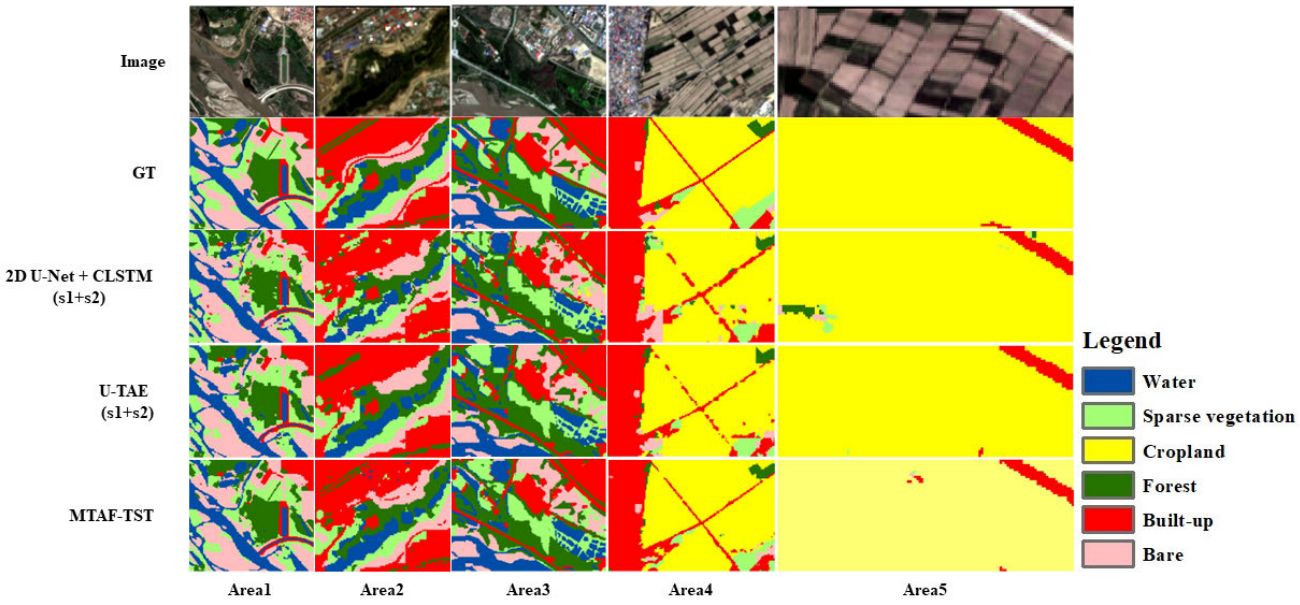


Fig. 11. Classification results on the Ili dataset. (Left to right) Five regions in the Ili dataset. (Top to bottom) Original image, GT, 2DU-Net + CLSTM(s1 + s2), U-TAE(s1 + s2) and MTAF-TST.

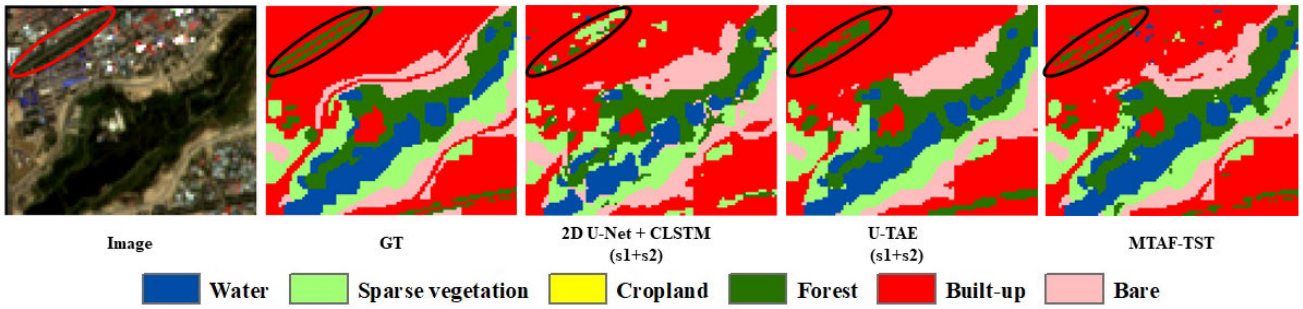


Fig. 12. Detail comparison on area 2.

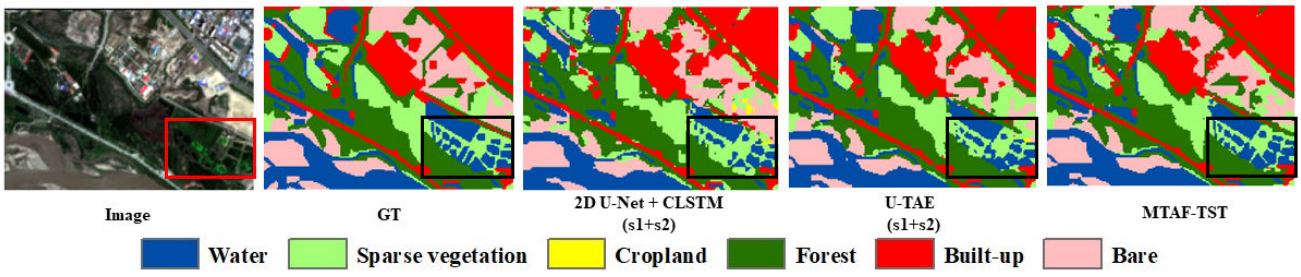


Fig. 13. Detail comparison on area 3.

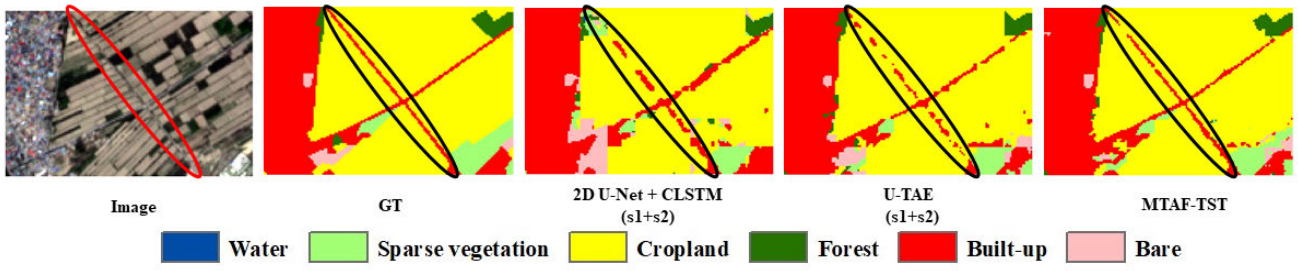


Fig. 14. Detail comparison on area 4.

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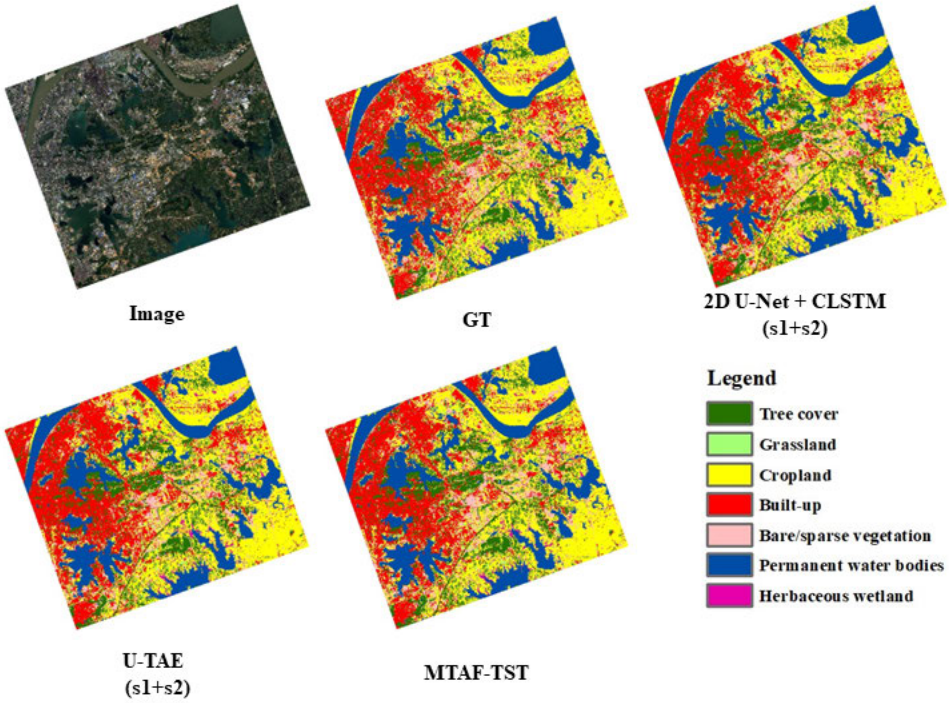


Fig. 15. Classification results on the Wuhan dataset. Image is the original remote sensing image, GT.

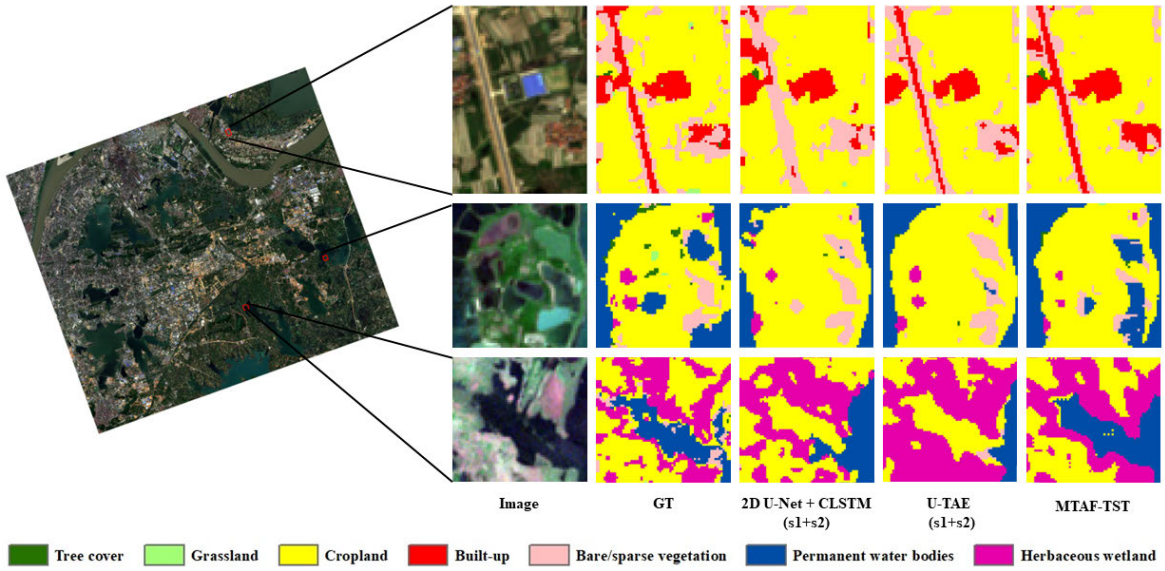


Fig. 16. Classification comparison of each model in three small regions on the Wuhan dataset.

based on TST. In the previous article, we have shown that MTAF-TST achieves the best results in comparative experiments. In order to verify the effectiveness of each part, in this section we gradually remove these two parts for ablation experiments. First, we remove the spatiotemporal information extraction module in the model to obtain the model Wo_TST and then remove the fusion module to obtain Wo_TST_MTAF. From Wo_TST_MTAF to Wo_TST, the model adds a data fusion module; from Wo_TST to MATF-TST, the model adds a spatiotemporal information extraction module. Fig. 18 shows the results of ablation experiments on two datasets. With the addition of the two modules, the classification accuracy of the model continues to increase, demonstrating the effectiveness of the two modules. From Fig. 19, we can see that adding the spatiotemporal information extraction module improves the accuracy more than adding the data fusion module, which also shows the importance of spatiotemporal information in the time series land cover classification task. Fig. 20 shows the magnitude of the change in different land cover accuracies before and after adding different modules. On the Ili dataset, adding the data fusion module mainly improves the accuracy of textured land covers such as water, sparse vegetation,

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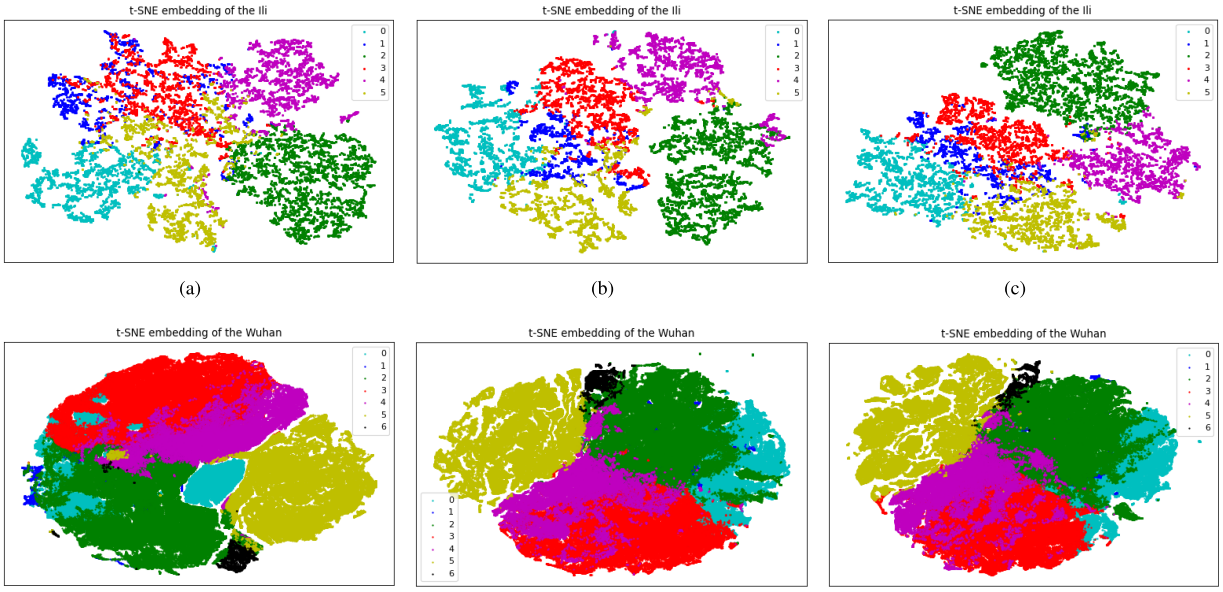


Fig. 17. t-SNE plots of the last layer features of all models. (a)–(c) Result of 2DU-Net + CLSTM(s1 + s2), U-TAE(s1 + s2), and MTAF-TST on Ili dataset. (d)–(f) Result on Wuhan dataset.

TABLE V
COMPARISON OF MODEL PARAMETERS

Model	Parameter	F1-score
WO_TST_MTAF	0.05M	70.36
WO_TST	0.07M	71.47
MTAF-TST	16M	75.22
2DU-Net+CLSTM(s1+s2)	50M	74.16
U-TAE(s1+s2)	21M	74.85

cropland, and built-up. At the same time, adding the spatiotemporal information extraction module mainly improves the accuracy of vegetation-related land covers such as sparse vegetation, cropland, forest, and bare. On the Wuhan dataset, adding the data fusion module mainly improves the accuracy of tree cover, grassland, bare/sparse vegetation, and herbaceous wetlands. Adding the spatiotemporal information extraction module improves accuracy for all land covers except built-up.

Therefore, the two modules can further consider spatiotemporal information on the basis of fusing the two kinds of data. This introduces texture information as well as phenological information to improve time-series land cover classification.

The parameter quantities and F1-score of the above three models are as shown in Table V. It can be seen that, the greater the number of modules, the greater the number of model parameters, and the higher the F1-score. In addition, the parameters of the comparison models are also counted. The results show that our proposed model can achieve better results with fewer parameters.

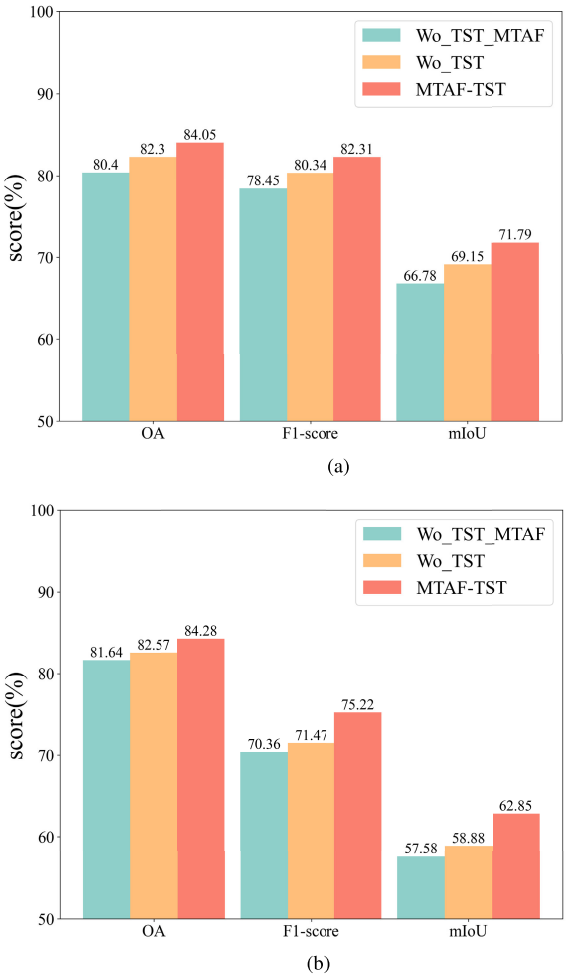


Fig. 18. Evaluation index scores of ablation experiments on the Ili and Wuhan datasets. (a) Ili dataset. (b) Wuhan dataset.

D. Discussion

By conducting experiments on the two datasets and analyzing the model structure and the experimental results, some main characteristics of the MTAF-TST model can be found.

The advantages of MTAF-TST are as follows.

1) MTAF can fully integrate the high-dimensional features of multisource data, which is conducive to utilizing

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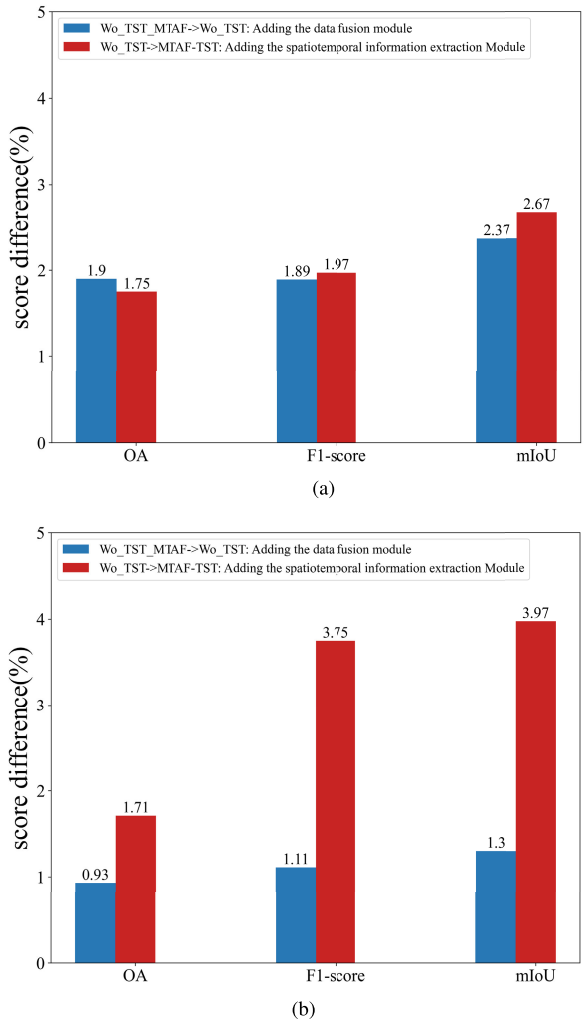


Fig. 19. Size of the increase in the score of each indicator before and after adding each module. (a) Ili dataset. (b) Wuhan dataset.

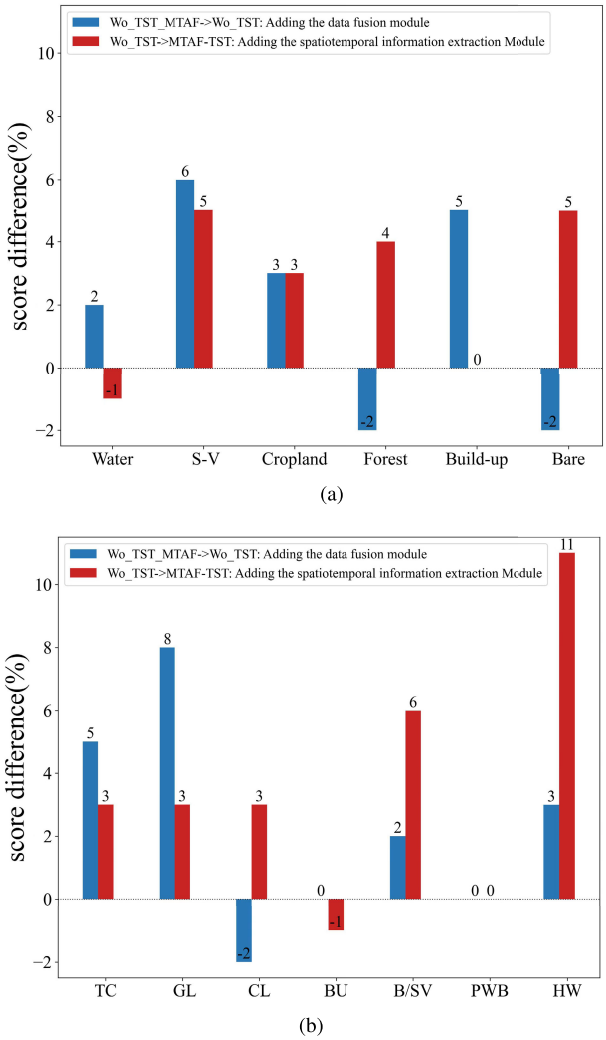


Fig. 20. Magnitude of the accuracy change for each land cover before and after adding each module. (a) Ili dataset. (b) Wuhan dataset.

the advantages of different data sources and improving the expressive ability of the model. TST can extract long-range temporal and spatial dependencies of SITS.

2) Feature-level fusion no longer directly concatenates latent representations of different branches. Instead, it fully considers the characteristics of time series data and introduces the time correlation between different data sources.
3) Comprehensively considering the losses of the three branches helps the model consider the classification preference of each type of data, improving the classification accuracy and robustness of the model.
4) Compared with other transformer-based land cover semantic segmentation models, MTAF-TST uses an attention mechanism, to extract the temporal evolution of ground objects using pixel time series, and extract the spatial correlation of ground objects phase by phase. This makes the extracted time and space dependencies independent of each other and does not affect each other, thereby improving the accuracy of land cover classification.

In addition, MTAF-TST also has the following shortcomings.

1) Since optical and SAR data are time series data, and each time phase has multiple spectra, the GPU memory usage is large.
2) When the MTAF module fuses multisource data, a one-time point pays attention to all the moments of the other source, but considering the greatest correlation with the surrounding time, MTAF can shorten the time span of attention.
3) The self-attention mechanism in TST has quadratic computational complexity [52].

V. Conclusion

To achieve high-precision time series land cover semantic segmentation, this study proposes a new deep-learning network architecture, MTAF-TST, which includes two main modules, MTAF and TST. The former can fuse SITS from different sensors; the latter can exploit the temporal and spatial context information of SITS to extract long-range spatiotemporal dependencies and then achieve high-precision time-series semantic segmentation. MTAF-TST can utilize the complementarity of multisource data and the synergy

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of spatiotemporal information to fully mine the space-time dependency and high-level semantic information of SITS and realize high-precision time series classification of land cover in urban areas with more frequent or enduring bad weather.

Unlike the previous direct concatenating of optical and SAR time series, MTAF can dynamically consider the correlation between multisource time series at different time points. However, in the fusion process of MTAF, each moment needs to pay attention to the data of all moments from other sources. In fact, the correlation between distant moments is not obvious. Therefore, only considering the neighbor moment dependence of a multisource time series will be a direction of improvement for MTAF. Furthermore, TST utilizes a self-attention mechanism to extract long-range dependencies, which fully considers the interaction between all moments and all spatial regions. It enhances the expressive power of the representative vectors of each moment or spatial region. However, due to the high computational complexity of the self-attention mechanism, introducing variants of self-attention to improve model performance will be another direction for research in the future.

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